

A COUNTEREXAMPLE TO THE CLASS 69 EQUIDISTRIBUTION CONJECTURE ON INVOLUTIONS

ABSTRACT. Fang, Fu, Kitaev, Li, Su, and Sun conjectured that the four length-two mesh patterns in Class 69 are equidistributed on involutions. We refute their conjecture, as stated in arXiv:2606.14367v1, at the minimal size $n = 3$. The two reverse-complement pairs have involution distribution polynomials $2 + q + q^2$ and $1 + 2q + q^2$, respectively; in particular, their avoidance counts are different.

1. THE COUNTEREXAMPLE

Mesh patterns were introduced by Brändén and Claesson [1]. Using the convention that a mesh cell is indexed by the coordinates of its lower-left corner, let $p_r = (12, R_r)$, where the four Class 69 patterns are listed in the order of Fang et al. [2, Conjecture 1]:

$$R_1 = \{(0, 0), (1, 1), (1, 2), (2, 1)\},$$

$$R_2 = \{(0, 1), (1, 0), (1, 1), (2, 2)\},$$

$$R_3 = \{(0, 2), (1, 0), (1, 1), (2, 1)\},$$

$$R_4 = \{(0, 1), (1, 1), (1, 2), (2, 0)\}.$$

Class 69 is the class of that name in [2, Remark 1.3], described there as the only remaining conjectural equidistribution class for mesh patterns of length two.

For completeness, let $p = (12, R)$ and $\sigma \in \mathfrak{S}_n$. Given $i < j$ with $\sigma_i < \sigma_j$, set

$$x_0 = 0, \quad x_1 = i, \quad x_2 = j, \quad x_3 = n + 1$$

and

$$y_0 = 0, \quad y_1 = \sigma_i, \quad y_2 = \sigma_j, \quad y_3 = n + 1.$$

The pair (i, j) is an occurrence of p if, for every $(a, b) \in R$, there is no k satisfying

$$x_a < k < x_{a+1} \quad \text{and} \quad y_b < \sigma_k < y_{b+1}.$$

Here an occurrence is indexed by the pair of positions (i, j) , whereas Fang et al. index it by the pair of values (σ_i, σ_j) ; since $i \mapsto \sigma_i$ is a bijection, the two conventions describe the same occurrences and give the same count. Write $\text{occ}_p(\sigma)$ for the number of occurrences. For

$$\mathcal{I}_n = \{\sigma \in \mathfrak{S}_n : \sigma^2 = \text{id}\},$$

2020 *Mathematics Subject Classification.* 05A05, 05A15.

Key words and phrases. mesh pattern, permutation pattern, involution, equidistribution, counterexample.

define

$$F_{n,p}^{\mathcal{I}}(q) = \sum_{\sigma \in \mathcal{I}_n} q^{\text{occ}_p(\sigma)}.$$

Equidistribution on involutions means equality of these polynomials for every n .

The conjecture we refute is stated in the concluding remarks of [2] and reads:

The patterns in the set $\{p_1, p_2, p_3, p_4\}$ are equidistributed on involutions. (The first two patterns, as well as the last two patterns, are trivially equidistributed via the composition of reverse and complement.)

In the source the four patterns are displayed as grid diagrams rather than named; substituting p_1, p_2, p_3, p_4 for those diagrams, in the order in which they appear there, is the only alteration we have made to the quoted sentence.

Theorem 1. *Conjecture 1 of Fang et al. [2], as stated in arXiv:2606.14367v1, is false. At $n = 3$,*

$$F_{3,p_1}^{\mathcal{I}}(q) = F_{3,p_2}^{\mathcal{I}}(q) = 2 + q + q^2,$$

whereas

$$F_{3,p_3}^{\mathcal{I}}(q) = F_{3,p_4}^{\mathcal{I}}(q) = 1 + 2q + q^2.$$

Proof. The involutions in $\mathfrak{S}_3$ are

$$\mathcal{I}_3 = \{123, 132, 213, 321\}.$$

For an increasing pair, the unique unused plot point lies in exactly one mesh cell, and the pair is an occurrence of p_r precisely when that cell is not in R_r .

For 123, the pairs (1, 2), (1, 3), and (2, 3) have unused-point cells (2, 2), (1, 1), and (0, 0), respectively. For 132, the pairs (1, 2) and (1, 3) have cells (2, 1) and (1, 2). For 213, the pairs (1, 3) and (2, 3) have cells (1, 0) and (0, 1). Finally, 321 has no increasing pair. Hence

	123	132	213	321
occ_{p_1}	1	0	2	0
occ_{p_2}	1	2	0	0
occ_{p_3}	2	1	1	0
occ_{p_4}	2	1	1	0

and summing the corresponding powers of q gives the stated polynomials. In particular, the avoidance counts are 2 for p_1 and p_2 , but 1 for p_3 and p_4 .

The separation is already visible in a coarser invariant. The seven increasing pairs listed above occupy seven distinct cells, namely every cell except (0, 2) and (2, 0). Hence R_1 and R_2 , which contain neither of these two cells, each block four of the seven pairs, while R_3 contains (0, 2) and R_4 contains

$(2, 0)$, so each blocks only three. The total occurrence counts $\sum_{\sigma \in \mathcal{I}_3} \text{occ}_{p_r}(\sigma)$, that is, the values $\frac{d}{dq} F_{3,p_r}^{\mathcal{I}}(q)|_{q=1}$, are therefore 3, 3, 4, 4. $\square$

Remark 2. The counterexample is minimal. For $n = 0$ and $n = 1$, no length-two pattern can occur. For $n = 2$, the involution 12 has one occurrence of every p_r , while 21 has none, so $F_{2,p_r}^{\mathcal{I}}(q) = 1 + q$ for all r .

Remark 3. Zhang and Zhao [3, Theorem 1.1] later proved the corresponding Class 69 equidistribution on the full symmetric groups $\mathfrak{S}_n$. The theorem above shows that this equivalence is not preserved after restriction to involutions.

REFERENCES

- [1] P. Brändén and A. Claesson, *Mesh patterns and the expansion of permutation statistics as sums of permutation patterns*, Electron. J. Combin. **18** (2011), no. 2, Paper P5, 14 pp., doi:10.37236/2001.
- [2] Q. Fang, S. Fu, S. Kitaev, H. Li, X. Su, and Z. Sun, *On mesh patterns of short length: Equidistribution and enumeration*, arXiv:2606.14367v1 [math.CO], 2026.
- [3] Z.-R. Zhang and H. Zhao, *The last distribution-equivalence class of mesh patterns of length 2*, arXiv:2607.04111v1 [math.CO], 2026.